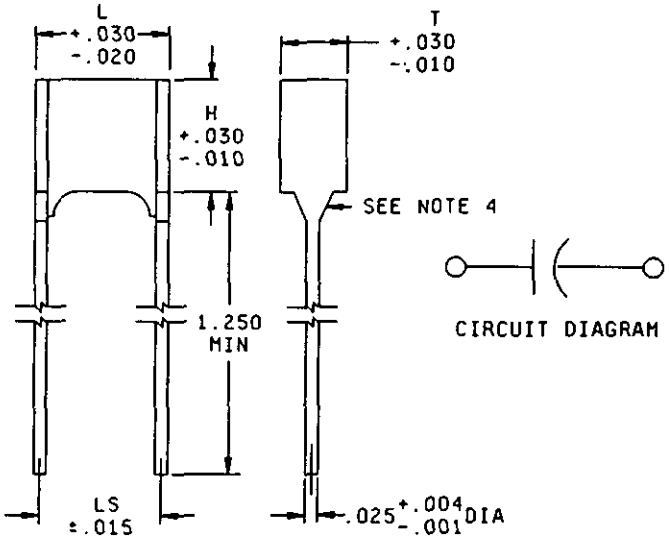
INCH-POUND
MIL-C-39022/11B(ER)
22 July 1993
SUPERSEDING
MIL-C-39022/11A(ER)
22 February 1990

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, METALLIZED PLASTIC DIELECTRIC, DIRECT-CURRENT, HERMETICALLY SEALED, STYLES CHR20, CHR21, CHR22, CHR23, CHR24, CHR25

This specification sheet is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39022.



Styles CHR20, CHR21, CHR22, CHR23, CHR24, CHR25

Inches	mm
.001	0.03
.004	0.10
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. The maximum solder meniscus on the leads shall be .050 (1.27 mm).
5. Lead length may be a minimum of 1 inch long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Capacitor configuration.

(B) denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case: Ceramic, hermetically sealed.

Dielectric: Polycarbonate, metallized.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Rated voltage: See table I for +85°C ratings. For +125°C operation, linearly derate +85°C rating by 50 percent.

Operating temperature range: -55°C to +125°C.

Failure rate (FR) level: M, P, R, S.

Preconditioning: Not applicable.

Flash point of impregnant: Not applicable.

Radiographic inspection (x ray): Not applicable.

Thermal shock: Method 107 of MIL-STD-202, condition B.

Seal: Method 112 of MIL-STD-202, test condition C, procedure IIIa, followed by test condition A. Leakage rate sensitivity 10^{-6} atm cm^3/s.

Dielectric withstanding voltage (DWV): In accordance with MIL-C-39022, except the charge time shall be 60 seconds minimum, 120 seconds maximum.

Insulation resistance (IR): In accordance with MIL-C-39022. When measured at the applicable test temperature and rated voltage, the insulation resistance shall equal or exceed the following values:

(B) Terminal to terminal:

Temperature	+25°C	+85°C	+125°C
Megohm x Microfarads	100,000	7,000	700
Except the insulation resistance in megohms need not exceed	200,000	70,000	7,000

(B) Terminal to case: Not applicable.

Capacitance: Method 305 of MIL-STD-202.

Test frequency: 1 kHz ±100 Hz.

Limit of accuracy: Within ±0.05 percent.

Dissipation factor (DF): Shall not exceed .2 percent at 1 kHz. Limit of accuracy shall be within ±0.05 percent.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet) 125 percent of dc rated voltage applied. See MIL-C-39022 for voltage limitations.

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(B) TABLE I. Electrical characteristics and dimensions.

Style	DC rated voltage	Capacitance value in µF	Capacitance tolerance	Dimensions 1/ T 2/	H 2/	L 2/	LS
CHR20	30	0.0330	F, G, J, K	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0390	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0470	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0560	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0680	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0820	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.1800	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.2200	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.2700	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.3300	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.3900	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.4700	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.5600	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.6800	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.8200	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	1.0000	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
CHR21	50	0.0100	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0120	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0150	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0180	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0220	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0270	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0330	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0390	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0470	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0680	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0820	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.1000	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.1200	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.1500	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.1800	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.2200	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.2700	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.3300	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.3900	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.4700	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.5600	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.6800	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
CHR22	100	0.0056	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0068	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0082	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0100	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0120	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0150	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0330	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0390	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0470	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0560	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0680	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0820	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.1000	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.1200	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.1500	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.1800	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.2200	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)

See footnotes at end of table.

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(B) TABLE I. Capacitor characteristics and dimensions - Continued.

Style	DC rated voltage	Capacitance value in µF	Capacitance tolerance	Dimensions 1/ T 2/	H 2/	L 2/	LS
CHR23	150	0.0033	F, G, J, K	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0039	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0047	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0056	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0068	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0220	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0270	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0330	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0390	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0560	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0680	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0820	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.1000	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
CHR24	200	0.0022	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0027	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0033	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0039	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0047	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0120	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0150	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0180	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0220	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0270	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0330	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0390	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0470	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0560	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.0680	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
CHR25	300	0.0010	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0012	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0015	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0018	"	.110(2.79)	.210(5.33)	.240(6.10)	.200(5.08)
"	"	0.0022	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0027	"	.170(4.32)	.270(6.86)	.240(6.10)	.200(5.08)
"	"	0.0033	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0039	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0047	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0056	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0068	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0082	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0100	"	.170(4.32)	.270(6.86)	.340(8.64)	.300(7.62)
"	"	0.0120	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0150	"	.190(4.83)	.380(9.65)	.340(8.64)	.300(7.62)
"	"	0.0180	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0220	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0270	"	.190(4.83)	.380(9.65)	.450(11.43)	.400(10.16)
"	"	0.0330	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)
"	"	0.0390	"	.220(5.59)	.500(12.70)	.450(11.43)	.400(10.16)

1/ Metric equivalents are in parentheses.
2/ T, H, and L are bare case dimensions (see figure 1). When specified on the purchase order, insulating covering for both "L" and "T" dimensions shall not exceed .025 inch (0.64 mm), nor shall the insulating covering for "H" dimension exceed .040 inch (1.02 mm).

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(B) Burn-in: DC: 48 hours at 125°C at 1.4 times the derated 125°C dc voltage rating. The current to each capacitor shall be limited to 1 ampere maximum by a series resistor.

AC: Not applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's), 50 percent of dc rated voltage applied.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: Five percent.

Immersion: Method 104 of MIL-STD-202, condition B.

DWV: Terminal to terminal: 200 percent of rated voltage.

IR: Terminal to terminal: Not less than 50 percent of value specified for insulation resistance.

Capacitance: Within ±1 percent of initial value.

DF: Not more than .3 percent.

Solderability: Method 208 of MIL-STD-202, the following details shall apply:

Number of terminations of each capacitor to be tested: Two.

Depth of immersion in flux and solder, both terminals shall be immersed to within .050 inch (1.27 mm) of the capacitor body.

Shock (specified pulse): Method 212 of MIL-STD-202, condition I, 50 percent of dc rated voltage applied. No arcing or other evidence of electrical breakdown.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, Capacitance, and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and C.

Condition A: Applied force 5 pounds.

Condition C: Applied force 5 pounds.

Low temperature: In accordance with MIL-C-39022.

Capacitance change with temperature (maximum):

At -55°C: -1.7 percent to -0.5 percent.

At +25°C (retrace): ±0.25 percent.

At +125°C: +1.2 percent to -0.7 percent.

Fungus: Method 508 of MIL-STD-810.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260°C ±5°C for 10 ±1 seconds).

IR: See initial value.

ΔCapacitance: Within ±0.25 percent of initial value.

DF: See initial value.

Resistance to solvents: Method 215 of MIL-STD-202.

ΔCapacitance: Maximum of ±0.25 percent.

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Life (at +100°C): Method 108 of MIL-STD-202.

Accelerated conditions: 140 percent of dc rated voltage for 2,000 +72, -0 hours.

Rated condition: 100 percent of dc rated voltage for 10,000 +96, -0 hours.

DF: (At +100°C) between 24 and 48 hours of test - see initial requirement.

IR: Terminal to terminal not less than 33.3 percent of initial requirement.

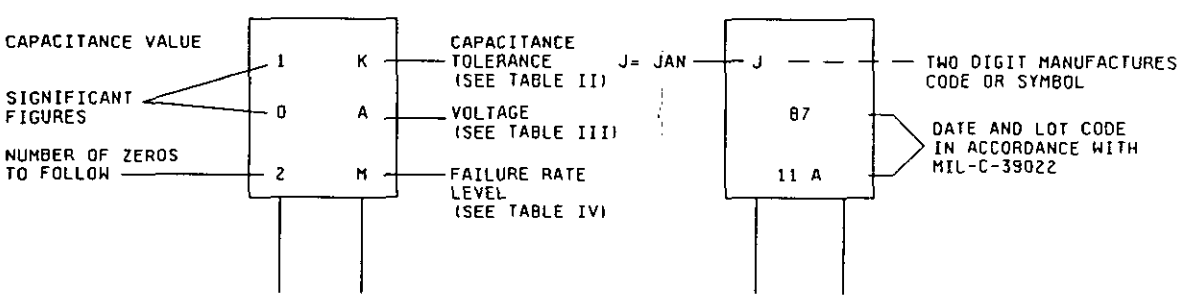
(8) The current to each capacitor will be limited to 1 ampere maximum by a series resistor.

Samples for complete qualification submission are as follows:

Style	Qualification group	Types to be submitted	Quantity for complete qualification groups
CHR20	30	M39022/11A333FM	38
CHR20	30	M39022/11A105FM	38
CHR21	50	M39022/11B103FM	38
CHR21	50	M39022/11B684FM	38
CHR22	100	M39022/11C562FM	38
CHR22	100	M39022/11C224FM	38
CHR23	150	M39022/11D332FM	38
CHR23	150	M39022/11D104FM	38
CHR24	200	M39022/11E222FM	38
CHR24	200	M39022/11E683FM	38
CHR25	300	M39022/11F102FM	38
CHR25	300	M39022/11F393FM	38

(8) Marking: Marking shall be in accordance with MIL-C-39022, except the Part or Identifying Number (PIN) example shall be M39022/11A102KM for bare case and M39022S11A102KM for sleeved units and minimum marking as shown in the following example will be permitted. The term PIN is equivalent to the term part number which was previously used in this specification. Full marking shall be included on the package.

EXAMPLE OF MINIMUM MARKING



Capacitance: The nominal capacitance value expressed in picofarads (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

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Capacitance tolerance: The capacitance tolerance in percent is identified by a single letter in accordance with table II.

TABLE II. Capacitance tolerance.

Symbol	Capacitance tolerance (±)
F	1 percent
G	2 percent
J	5 percent
K	10 percent

Rated voltage: The rated voltage is identified by a single letter in accordance with table III.

TABLE III. Rated voltage.

Symbol	DC rated voltage
A	30
B	50
C	100
D	150
E	200
F	300

Failure rate level: The failure rate level in percent per 1,000 hours is identified by a single letter in accordance with table IV, and is based on rated voltage at +100°C.

TABLE IV. Failure rate level.

Symbol	Failure rate level (percent per 1,000 hours)
M	1.0
P	0.1
R	0.01
S	0.001

Substitutability for capacitance tolerance shall be as follows:

Capacitance tolerance	Will replace capacitance tolerance
F	G, J, K
G	J, K
J	K
K	

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Substitutability for voltage shall be as follows:

Volts dc	Will replace voltages
300	200,150,100,50,30
200	150,100,50,30
150	100,50,30
100	50,30
50	30
30	

Substitutability for failure rate levels shall be as follows:

Failure rate level	Will replace failure rate levels
S	R, P, M
R	P, M
P	M
M	

CONCLUDING MATERIAL

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-A471)

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